Gaussian pentagon matrix elements: a certified heat cutoff and unresolved quadrature

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Abstract

For the specified Kim–Scarinci phase at $\hbar = 1$, a punctured H^1 approximation bounds the heat-cutoff error in a Gaussian pentagon matrix element by 0.000982 at $\epsilon = 10^{-10}$. An independent integer/rational enclosure certifies the numerical constant. Actual FFT and Hermite evaluations give nonzero discrete matrix elements, but their quadrature errors remain uncertified. A flat control shows why damping-induced residuals alone cannot refute the undamped identity. No continuous counterexample or physical holographic conclusion is established.

1 The precise scope of the certificate

Use dimensionless coordinates, $d\mu = dx\,dy/(2\pi^2)$ and

$$P_j = -\pi i \partial_j, \quad A = F_1(X, Y), \quad B = F_1(P_y, P_x), \quad M = e^{iXY/\pi},$$

where F_1 is the bounded paired phase of KS [1], defined almost everywhere by

$$F_1(x, y) = \Phi^i(x + iy)/\overline{\Phi^i(x + iy)}.$$

The identity being reproduced is $D = AB - BM^{-1}BMA = 0$. Let $H_\epsilon = e^{-\epsilon(P_x^2+P_y^2)}$, $B_\epsilon = BH_\epsilon$, and $D_\epsilon = AB_\epsilon - B_\epsilon M^{-1}B_\epsilon MA$. The bounded Fourier-kernel formulation uses the Gaussian-group framework [2]; it does not split unbounded chiral factors.

For $f = \sqrt{2}\, e^{-(x^2+y^2)/(2\pi)}$, normalized in $L^2(\mu)$, put $v = MAf$ and $E_\epsilon(u) = \|(1 - H_\epsilon)u\|$. The preceding bounded-word estimate gives

$$|\langle f, (D - D_\epsilon)f\rangle| \le 2E_\epsilon(f) + E_\epsilon(v), \qquad E_\epsilon(f)^2 = \frac{2(\pi\epsilon)^2}{(1 + \pi\epsilon)(1 + 2\pi\epsilon)}. \tag{1}$$

Here we bound the last term explicitly. All constants are for $\hbar = 1$ and this fixed test. The certificate concerns (1), not numerical integration of $\langle f, D_\epsilon f\rangle$.

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2 A derivative bound on a punctured box

Set $q = e^{-\pi}$ and $a_j = (2j+1)\pi$. The known product formula gives

$$F_1(x,y) = \prod_{j\geq 0} g_j(x,y)g_j(y,x), \qquad g_j(x,y) = \frac{1+e^{x-a_j-iy}}{1+e^{x-a_j+iy}}. \tag{2}$$

Each factor has modulus one. For $z = e^{x-a_j+iy}$ a local phase obeys

$$|\nabla \arg g_j| = 2\left|\frac{z}{1+z}\right|.$$

Write $u = x - a_j$ and let $\tau \in [-\pi, \pi]$ be the angular displacement from an odd multiple of π. Then

$$\begin{aligned}\left|\frac{z}{1+z}\right|^2 &= \frac{e^u}{2(\cosh u - \cos\tau)} \leq 1 + \frac{1}{\cosh u - \cos\tau} \\ &\leq 1 + \frac{\pi^2}{2(u^2+\tau^2)}.\end{aligned} \tag{3}$$

The last step uses $\cosh u - 1 \geq u^2/2$ and $1 - \cos\tau \geq 2\tau^2/\pi^2$.

On $[-9,9]^2$, all $j \geq 1$ terms are uniformly controlled by

$$T_{\text{prod}} = \frac{e^{9-3\pi}}{(1-e^{9-3\pi})(1-e^{-2\pi})}.$$

Their radial moduli are less than one, so the geometric denominator estimate applies term by term. Let d_x be the distance to the nearest $(\pi, (2k+1)\pi)$, and d_y to the nearest $((2k+1)\pi, \pi)$. A local phase $\theta = \arg(MF_1)$ satisfies

$$|\nabla\theta| \leq C + \frac{\sqrt{2}\pi}{d_x} + \frac{\sqrt{2}\pi}{d_y}, \qquad C = \frac{9\sqrt{2}}{\pi} + 4 + 4T_{\text{prod}}.$$

Consequently

$$|\nabla\theta|^2 \leq 3C^2 + 6\pi^2(d_x^{-2} + d_y^{-2}). \tag{4}$$

Only the odd ordinates $\pm\pi, \pm 3\pi$ are needed on this box. These estimates remove more points than are singular in the combined phase, which is harmless for an upper bound. In particular, no uncancelled singularity at (π, π) is asserted.

3 The punctured H^1 approximation

Let $a(s)$ equal one on $[-8,8]^2$, taper linearly in each coordinate to zero at $|x| = 9$ or $|y| = 9$, and vanish outside. Use holes of radius $\rho = 10^{-4}$ around the seven points

$$\mathcal{P} = \pi\{(1,-3),(1,-1),(1,1),(1,3),(-3,1),(-1,1),(3,1)\}.$$

At each point take a radial cutoff that is zero up to ρ, linear up to one at 2ρ, and one thereafter. Their transition annuli are disjoint. Let b be their product and $w = abv$. It is compactly supported and in H^1. The probability density of $|v|^2 d\mu$ in ordinary coordinates is $\varrho(s) = \pi^{-2} e^{-|s|^2/\pi}$. Define

$$K(d,R) = \pi^{-2} e^{-(d-R)^2/\pi}, \quad T = \frac{1}{4} e^{-64/\pi}, \quad J_\rho = \sum_{p\in\mathcal{P}} 4\pi\rho^2 K(|p|, 2\rho).$$

The Gaussian union-tail bound gives mass outside $[-8,8]^2$ at most T, so $\|v - w\| \le \sqrt{T + J_\rho}$. Indeed, bound $\int_L^\infty e^{-x^2/\pi} dx$ by $\int_L^\infty (x/L) e^{-x^2/\pi} dx$; see also [3].

For $R = 1/2$, $c = 3/5$, $d = |p|$, the inverse-distance integral outside the hole is bounded by

$$I_p = 2\pi K(d,R) \log(R/\rho) + \frac{1}{(1-c)^2 d^2} + \frac{e^{-c^2 d^2/\pi}}{R^2}. \tag{5}$$

The first term integrates $\rho < |s-p| < R$. For the remainder split at $|s| = cd$: inside, $|s-p| \ge (1-c)d$; outside, the radial Gaussian tail is $e^{-c^2 d^2/\pi}$. Let $\mathcal{P}'$ be the list $\mathcal{P}$ with (π,π) repeated once, accounting for both product orientations. Equation (4) and the real Gaussian amplitude derivative give

$$G = \sqrt{\frac{2}{\pi} + 3C^2 + 6\pi^2 \sum_{p\in\mathcal{P}'} I_p} + \sqrt{2T} + \sqrt{\sum_{p\in\mathcal{P}} 4\pi K(|p|, 2\rho)} \;\ge\; \|\nabla w\|. \tag{6}$$

The last two terms bound the square cutoff and hole cutoffs. Since $\|1 - H_\epsilon\| \le 1$ and $(1 - e^{-z})^2 \le z$ for $z \ge 0$,

$$E_\epsilon(v) \le \sqrt{T + J_\rho} + \pi\sqrt{\epsilon}\, G. \tag{7}$$

Thus the original phase need not lie in H^1.

4 Elementary enclosure and a direct left-word integral

Combining (1) and (7) gives

$$\mathcal{B}_\epsilon = 2\sqrt{\frac{2(\pi\epsilon)^2}{(1+\pi\epsilon)(1+2\pi\epsilon)}} + \sqrt{T + J_\rho} + \pi\sqrt{\epsilon}\, G. \tag{8}$$

Two independent evaluations are saved: 40/70-digit `mpmath.iv`, and a Python integer/Fraction implementation. Since interval support in mpmath is documented as experimental [4], the final strict inequalities also use the latter.

The rational implementation rounds outward to 120-bit dyadic endpoints after arithmetic. It uses the alternating Machin formula $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ with next-term tails. For $\exp(x)$, range reduction gives $0 \le x \le 1/8$; a degree-24 Taylor sum has tail at most its next term divided by $1 - x/26$. Negative arguments use reciprocals. For $\log y$, powers of two reduce y to $[1, 2]$, then

$$\log y = 2\sum_{j=0}^{39} \frac{z^{2j+1}}{2j+1} + R, \quad z = \frac{y-1}{y+1}, \quad 0 \le R \le \frac{2z^{81}}{81(1-z^2)}.$$

Integer square roots enclose square roots. The final comparisons are exact rational endpoint comparisons:

ϵ	approximate $\mathcal{B}_\epsilon$	certified strict upper
10^{-8}	.009631522192	.01
10^{-10}	.000981734116	.000982
10^{-12}	.000116763225	.000117

This is an arithmetic certificate for the stated bound, not a proof-assistant verification of all software or of the pentagon.

An independent continuous integral for the left word is

$$\langle f, AB_\epsilon f\rangle = \frac{2}{1+2\pi\epsilon}\,\mathbb{E}\Big[F_1(S)F_1(T)e^{i(S_xT_y+S_yT_x)/\pi}\Big], \tag{9}$$

where S, T are independent centered planar Gaussians with covariances πI and $\pi I/(1+2\pi\epsilon)$. This follows directly from the Fourier transform of f. Four-dimensional tensor Gaussian quadrature is evaluated through two separable matrix products, at orders 128–768. It uses neither periodic FFT nor a truncated Hilbert basis. For $F_1 = 1$, (9) is exactly $(1+\pi\epsilon)^{-1}$.

5 Actual regulated matrix elements and controls

We evaluated the same D_ϵ with periodic FFT and oversampled Hermite projection. The heat multiplier is included at both B occurrences. Heat-induced norm loss is not counted as unitary projection leakage. Thirty-seven records retain both complex word values, their difference, norms, parameters and scalar product tails.

For $\hbar = 1$ and $\epsilon = 10^{-10}$, representative defects are:

method/settings	real part	imaginary part
FFT 1024, $L = 20$	.00391212915	-.00431820856
FFT 2048, $L = 20$	.00391230420	-.00431827530
FFT 2048, $L = 28$	.00396855893	-.00434478257
Hermite 64, $q = 256$	.00391447296	-.00424922552
Hermite 128, $q = 512$	.00390554046	-.00415910020
Hermite 256, $q = 768$	.00392158989	-.00415995798

Their moduli are around .0057–.0059. At fixed L, doubling FFT size changes the answer much less than changing L. It primarily extends the momentum range while its momentum spacing stays π^2/L; apparent convergence in size alone cannot control periodization and momentum quadrature.

For the left word alone, direct order-768 quadrature at 10^{-10} gives

$$.914113080399 + .000104730439\,i.$$

FFT 2048, $L = 20$ gives $.914141192384 + .000103846566\,i$, and Hermite 256, $q = 768$ gives $.914061270561 + .000098283811\,i$. These differences have not been enclosed by rigorous error bounds. Direct quadrature with $F = 1$ agrees with the exact Gaussian formula within 10^{-12}, and its flat left-word control agrees with FFT within 10^{-12}; these controls do not certify the singular paired-phase integrals.

The flat control is essential. For $F_0(x, y) = \exp[-iy \log(1 + e^x)/\pi]$, the undamped identity is known [1]. FFT and Hermite give undamped scalar residuals below 10^{-12}. Yet at $\epsilon = .01$ both give a defect approximately .0334367989605. At $\epsilon = 10^{-10}$ it decreases to $3.6133\,10^{-10}$. Damping does not preserve the exact pentagon. A nonzero regulated defect without a removal-error estimate proves nothing about D.

6 What is established and what is missing

The heat-removal part of the Gaussian matrix-element error is now explicitly bounded. At $\epsilon = 10^{-10}$ it is below .000982, including puncture, spatial-tail, derivative and arithmetic contributions. This improves the earlier empirical phase-loss estimate without assuming H^1 regularity for MAf.

The actual regulated words have also been evaluated in two discretizations, and their left word has an independent continuous-kernel quadrature. The results motivate a more precise test, but none supplies a certified enclosure for $\langle f, D_\epsilon f\rangle$. It is invalid to subtract .000982 from the modulus of a tabulated approximation and call the result a continuous lower bound.

The next required calculation is a rigorous enclosure

$$\langle f, D_\epsilon f\rangle \in \mathcal{I}_\epsilon.$$

It must account for scalar product truncation, floating-point rounding, Fourier/physical tails, phase neighborhoods and quadrature. A true nonzero witness would require

$$\inf_{z\in\mathcal{I}_\epsilon} |z| > \mathcal{B}_\epsilon.$$

Failure to meet this inequality is inconclusive, not proof of the identity. The current interval is missing.

Prioritize a singularity-aware enclosure of the right word, which remains the more difficult integral. The flat and unit controls, actual complex word values and failed precision targets must remain visible. Further unbounded-factor analytic continuation is not needed for this bounded test. Literature may still provide an exact kernel that simplifies the integration.

The stage contains 37 matrix rows, 12 direct-left rows, six interval evaluations and three rational certificates. The 188 replay checks verify saved arithmetic, controls, hashes and exact certificate regeneration. They do not certify the discrete matrix values. This is a model calculation in a fixed Hilbert representation, not an observation, a complete finite-G construction or a no-go result for de Sitter holography.

References

[1] H. K. Kim and C. Scarinci, *A quantization of moduli spaces of 3-dimensional gravity*, 2112.13329v2, §6.3 and Proposition 6.15.

[2] S. Garoufalidis and R. Kashaev, *A meromorphic extension of the 3D index*, 1706.08132v2, Appendix B. The preceding paired-kernel note gives the full dictionary.

[3] NIST Digital Library of Mathematical Functions, §7.8, Gaussian tail inequalities. We use the elementary weaker bound derived in the text.

[4] mpmath 1.3.0 documentation, Contexts: interval arithmetic, accessed 1 November 2026.